Module 2 Cheat Sheet – Introduction to Linux Commands

Getting Information

Return your user name:

```
None
whoami
```

Return your user and group ID:

```
None
id
```

Return operating system name, username, and other info:

```
None
uname -a
```

Display reference manual for a command:

```
None
man top
```

List available `man` pages with descriptions:

```
None
man -k .
```

Get help on any command (e.g., curl):

```None
curl --help
```

This provides a brief overview of the command's usage and options.

Return the current date and time:

```None
date
```

Navigating and Working with Directories

List files and directories by date (newest last):

```None
ls -lrt
```

Find files in directory tree that end in `.sh`:

```None
find -name "*.sh"
```

Return path to present working directory:

```None
pwd
```

Create a new directory:

```None
mkdir new_folder
```

Change current directory:

- Up one level:

```None
cd ../
```

- To home:

```None
cd ~
```

or

```None
cd
```

- To another directory:

```None
cd path_to_directory
```

Remove directory (verbose):

```None
rmdir temp_directory -v
```

Monitoring System Performance and Status

List running processes:

```None
ps
```

List all processes:

```None
ps -e
```

Display resource usage:

```None
top
```

List mounted file systems and usage:

```None
df
```

Creating, Copying, Moving, and Deleting Files

Create or update a file:

```None
touch a_new_file.txt
```

Copy a file:

```None
cp file.txt new_path/new_name.txt
```

Move or rename a file:

```None
mv this_file.txt that_path/that_file.txt
```

Remove a file (verbose):

```None
rm this_old_file.txt -v
```

Working with File Permissions

Change/modify file permissions to 'execute' for all users:

```None
chmod +x my_script.sh
```

Change/modify file permissions to 'execute' only for you, the current user:

```None
chmod u+x my_file.txt
```

Remove 'read' permissions from group and other users:

```None
chmod go-r
```

Displaying File and String Contents

Display file contents:

```
None
cat my_shell_script.sh
```

Display file page-by-page:

```
None
more ReadMe.txt
```

Show first 10 lines:

```
None
head -10 data_table.csv
```

Show last 10 lines:

```
None
tail -10 data_table.csv
```

Display text or variables:

```
None
echo "I am not a robot"
echo "I am $USERNAME"
```

Basic Text Wrangling

Sorting and Removing Duplicates

Sort and display lines of file alphanumerically:

```None
sort text_file.txt
```

Sort in reverse:

```None
sort -r text_file.txt
```

Drop consecutive duplicated lines and display result:

```None
uniq list_with_duplicated_lines.txt
```

Displaying basic stats:

Count lines:

```None
wc -l table_of_data.csv
```

Count words:

```None
wc -w my_essay.txt
```

Count characters:

```None
wc -m some_document.txt
```

Extracting lines of text containing a pattern:

Some frequently used options for `grep`

Common options:

Option	Description
-n	Print line numbers along with matching lines
-c	Get the count of matching lines
-i	Ignore the case of the text while matching
-v	Print all lines which do not contain the pattern
-w	Match only if the pattern matches whole words

Extract lines containing the word "hello", case insensitive and whole words only:

```
None
grep -iw hello a_bunch_of_hellos.txt
```

Extract lines containing the pattern "hello" from all files in the current directory ending in .txt:

```
None
grep -l hello *.txt
```

Merge two or more files line-by-line, aligned as columns:

Suppose you have three files containing the first and last names of your customers, plus their phone numbers.

Use `paste` to align file contents into a Tab-delimited table, one row for each customer:

```
None
paste first_name.txt last_name.txt phone_number.txt
```

Use comma delimiter:

```
None
paste -d "," first_name.txt last_name.txt phone_number.txt
```

Use the `cut` command to extract a column from a table-like file:

Suppose you have a text file whose rows consist of first and last names of customers, delimited by a comma.

Extract first names, line-by-line:

```
None
cut -d "," -f 1 names.csv
```

Extract the second to fifth characters (bytes) from each line of a file:

```
None
cut -b 2-5 my_text_file.txt
```

Extract the characters (bytes) from each line of a file, starting from the 10th byte to the end of the line:

```
None
cut -b 10- my_text_file.txt
```

Compression and Archiving

Archive a set of files:

```
None
tar -cvf my_archive.tar.gz file1 file2 file3
```

Compress files:

```None
zip my_zipped_files.zip file1 file2
zip my_zipped_folders.zip directory1 directory2
```

Extract files from a compressed zip archive:

```None
unzip my_zipped_file.zip
unzip my_zipped_file.zip -d extract_to_this_directory
```

Working with networking commands

Print hostname:

```None
hostname
```

Send packets to URL and print response:

```None
ping www.google.com
```

Display or configure system network interfaces:

```None
ip
```

Display contents of file at a URL:

```None
curl <url>
```

Download file from a URL:

```None
wget <url>
```
